AeroSense short user feedback form

Revision: 2026-10-13 R4. Planned duration: 5–10 minutes.

Participant code: _____ Date: _____ Manual version: _____ Observer code: _____

This session identifies problems in the manual and interface. It is not a test of your ability. State unclear steps. You can stop at any time. Do not enter names, telephone numbers, student IDs, or other identifying details.

Data use: ☐ I agree to anonymous summaries of task times, errors, and comments in the team's iGEM project records.

Separate image choice: ☐ Hands and parts only; no face ☐ No photos or video. Declining images does not prevent participation.

Preparation and scope

The developer prepares dry, cell-free, unpowered empty-well demonstration parts. The developer opens DEMO on a computer. Prepare the URL, connection settings, user manual, and download location. Participants do not flash firmware, connect power, culture cells, or handle samples.

DEMO tests interface tasks only. The developer separately completes physical-board, standard-liquid, and required functional validation. This form cannot replace those results.

Involved in design: ☐ No ☐ Yes. Operation experience: ☐ None ☐ Some ☐ Familiar.

Session: ☐ First attempt ☐ Retest after revision. UI mode is DEMO throughout.

Task observations observer completes

Provide the same manual and prepared interface. Do not demonstrate the solution first. Include reading in task time. Record every prompt. If damage is possible, stop the task and record the cause.

Use independent completion, assisted completion, or incomplete. Mark omitted tasks as untested. Record actual time if tasks need longer. Do not invent completion to fit ten minutes.

Task	Seconds	Outcome	Prompts	Difficulty or direct comment
1 Identify empty-well orientation and replacement without contact with optical surfaces				
2 Open the prepared UI, connect to DEMO, and identify its label				
3 Set 1 minute and 3 measurements, find Pause, then start				
4 Pause the demonstration, verify Paused, then end the schedule				
5 Export JSON. Find and open the downloaded file.				

DEMO mistaken for measurements: ☐ No ☐ Yes ☐ Not observed.

Relevant manual page/figure: _____ Total time: _____ seconds.

Your ratings

1 strongly disagree; 2 disagree; 3 neutral/uncertain; 4 agree; 5 strongly agree. Select N/A for unperformed tasks. This is a project-specific form, not SUS.

Statement	1	2	3	4	5	N/A
Empty-well orientation and replacement were clear.	☐	☐	☐	☐	☐	☐
I could identify DEMO and distinguish it from measurements.	☐	☐	☐	☐	☐	☐
I could find Start, Pause and End schedule.	☐	☐	☐	☐	☐	☐
I could export and find the JSON file.	☐	☐	☐	☐	☐	☐
I could complete the specified tasks with the manual.	☐	☐	☐	☐	☐	☐

1. Which step was least clear? _____

2. If one thing could change, what should change first? _____

Team changes and retest after the participant leaves

Prioritize one or two problems. Preserve actual evidence. Do not assign favorable results to unperformed tasks.

Problem and evidence	Actual change, file/page/date	Same-task retest result

Retest participant code: _____ Same person: ☐ Yes ☐ No.

Retested task: _____ Seconds: _____ Prompts: _____.

Unresolved problem: _____

Publish individual task outcomes and times. Add medians if useful. Explain the small sample and learning effects in repeated participants. This short trial assesses empty-well identification and UI operation. It does not establish complete assembly usability or cell function.